

PROG7311

GLMS

Portfolio Of Evidence (Final part)

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Report

Why Automated Testing is Critical in a CI/CD Pipeline

Automated testing lets developers verify that new features and bug fixes do not introduce regressions before code reaches production (Testlio, 2024). Each commit triggers the test suite, so defects are caught at the point of introduction rather than during later integration (Testlio, 2024). As release frequency increases, teams that rely on manual regression testing tend to shortcut the process, which allows bugs to reach end users (Verma, 2018).

Why Docker Solves the “It Works on My Machine” Problem

A Docker container packages an application together with all its dependencies, so the same image runs identically in development, test, and production (Docker Inc., 2026). Developers write code locally using containers, push those containers to a test environment for automated and manual testing, and promote the same image to production when testing is complete (Docker Inc., 2026). Because the running unit never changes between stages, environment-specific configuration differences cannot cause divergent behaviour, so the “it works on my machine” problem does not arise (Docker Inc., 2026).

github link: <https://github.com/EMWCCN/prog7311-poe-part3-UR91K>

References

Docker Inc. (2026) *What is Docker?*. Available at: <https://docs.docker.com/get-started/docker-overview/> (Accessed: June 9, 2026).

Testlio (2024) *Automation for CI/CD*. Available at: <https://www.testlio.com/blog/ci-cd-test-automation> (Accessed: June 9, 2026).

Verma, A. (2018) *Why Test Automation is Essential for CI/CD*. Available at: <https://www.functionize.com/blog/why-test-automation-is-essential-for-ci-cd> (Accessed: June 9, 2026).

Appendix A - Disclaimer

This paper was typeset using Typst (typst.app), a markup-based typesetting system. The result may look more formally typeset than a typical Word submission; that is the nature of Typst. The source file is available on request.